

Problem Set - 4

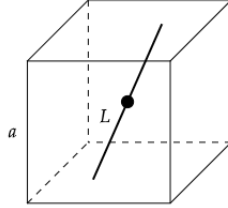
Kutay Kulbak

30 Eylül 2024

1 IdPhO 2020 - Short Problems

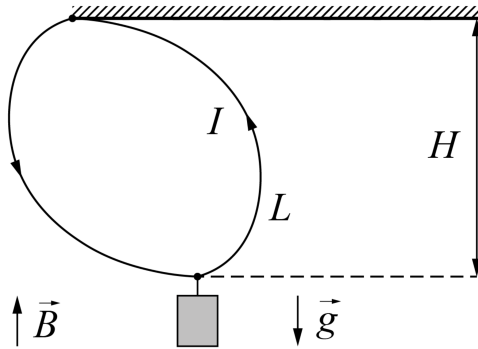
1.1 Cubic Oscillations

A narrow straight channel passes through the center of a fixed cube with a side a . The cube is uniformly charged, the charge density is ρ . The distance from the cube center to the point of intersection of the channel and a face is L . In the channel there is a particle of a mass m and a charge q . Find the period of small oscillations of the particle near the center. The gravitational interaction of the particle and the cube can be neglected. The cube and the particle are oppositely charged.



1.2 Suspension in Magnetic Field

A current I flows through a loop made of a weightless flexible wire. The loop upper point is attached to the ceiling and a weight is suspended to its lowest point. The half length of the loop is L . The loop is placed in a vertical magnetic field B . The system has reached a stable equilibrium in which the point of suspension at the ceiling and the point of weight suspension are not on the same vertical. Find the wire tension T and the weight P if the distance from the ceiling to the lowest point of the loop is H .



1.3 Rod in Magnetic Field

A weightless rod of a length $2R$ is placed perpendicular to a uniform magnetic field $\vec{B}$. Two identical small balls of mass m and charge q each are attached at the rod ends. Let us direct z -axis along the magnetic field and place the origin at the rod center. The balls are given the same initial velocity v but in opposite directions so that one of the velocities is precisely in the z -direction. What are the maximum coordinates z_{max} of the balls? Express your answer in terms of q , B , m , v , and R . Find the magnitude of the ball accelerations at this moment and express your answer in terms of q , B , m , v , R , and z_{max} .

2 Mechanics

2.1 Particle in a Sphere

A particle of mass m is currently undergoing circular motion at angular frequency $\omega_0 > \sqrt{g/R}$ in the interior of a massive sphere of radius R . Let θ be the angle between the vertical axis, passing through the center of the sphere and pointing downwards, and the position of the particle in spherical coordinates, taking a positive value in the anti-clockwise direction. Suppose that the particle is given a slight push in the θ direction, determine the angular frequency of small oscillations in the θ direction. (I forgot this one too lmao)

2.2 Colliding Balls

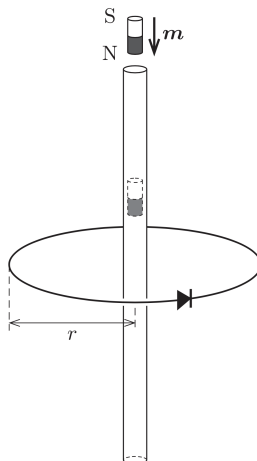
Two small balls are threaded onto a frictionless horizontal rod that protrudes from a vertical wall. The lighter ball with mass m is initially at rest, a distance L from the wall, while the much heavier second ball, of mass M , approaches the wall from a distance greater than L (*see figure*). After their elastic collision, the ball of mass m slides towards the wall, bounces back elastically, and again collides with the heavier ball. The process then repeats itself, over and over. How close does the heavier ball get to the wall? Consider the balls as point-like particles, and, where appropriate, assume that $m \ll M$. (200 More, P44)



3 Electromagnetism

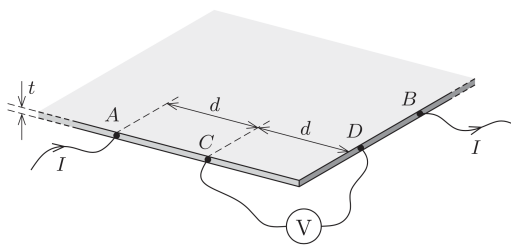
3.1 Falling Magnet

A closed circular loop of radius r is made from a wire of resistance R and a diode, which can be considered ideal. The loop is held in a horizontal plane, and a long, vertical glass tube passes through its centre (*see figure*). Find the charge that flows through the diode if a small bar magnet with magnetic moment $\vec{m}$ falls through the tube. (200 More, P180)



3.2 Resistivity of Plate

We aim to measure the resistivity of the material of a large, thin, homogeneous square metal plate, of which only one corner is accessible. To do this, we chose points A , B , C and D on the side edges of the plate that form the corner (*see figure*). Points A and B are both $2d$ from the corner, whereas C and D are each a distance d from it. The length of the plate's sides is much greater than d , which, in turn, is much greater than the thickness t of the plate. If a current I enters the plate at point A , and leaves it at B , then the reading on a voltmeter connected between C and D is V . Find an expression for the resistivity ρ of the plate material. (200 More, P174)



4 Optics

4.1 Weird Planet

On a spherical planet, the refractive index of the atmosphere, as a function of altitude h above the surface, varies according to the formula

$$n(h) = \frac{n_0}{1 + \varepsilon h}$$

where n_0 and ε are constants. Curiously, any laser beam, directed horizontally, but at an arbitrary altitude, follows a trajectory that circles the planet. What is the radius of the planet?

5 Miscellaneous

5.1 Titanium-devouring Little Green Men

A space ship carrying titanium-devouring little green men has found and landed on a perfectly spherical planet of radius R . A narrow trial shaft has been bored from point A on the surface of the planet to O , its centre; this has confirmed that the whole planet is made of homogeneous (edible) titanium. In addition, according to the measurements made, the temperature inside the narrow shaft is constant, and equal to T_0 . The planet has an atmosphere with molar mass M , and the atmospheric pressure at its surface is p_A .

a) Find the air pressure at the bottom of the shaft.

After the exploratory drilling, work has continued, and the little green men have started secret excavation of the titanium, as a result of which they have formed a spherical cavity of diameter AO inside the planet, as illustrated in the figure. The excavated titanium is being transported away using expendable cargo space craft. Air from the atmosphere has moved to fill the cavity and, as a consequence, the pressure at the access point A has decreased from p_A to p'_A .

b) Assuming that the temperature everywhere inside the cavity is the same as it was in the shaft, how has the atmospheric pressure at O changed? (200 More, P32)

